

An IEEE 1451 Vehicular Sensor Network Development

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Abstract—The mobility platform proposed is based on e-bikes and a set of charging stations, spread all over the city, that use the sun's energy to charge the batteries of e-bikes. At the same time, the mobility of e-bikes is used to collect environmental parameters related to air quality. The user's mobile phone is used to connect the e-bike to the traceability and data collection system. The merge of technologies such as IoT and Sensor Networks (SN) in vehicles have originated the concept of Vehicular SN (VSN). To promote the acceptance and implementation of standardized SNs, the objective of the present work is to present a VSN compliant with the IEEE 1451 family of standards. It is presented the overall architecture of the VSN and is it described the implementation of services, methods and group of transducers to achieve the expected smartness of a VSN. Also, will be underlined how all VSN components interoperate as well as what operations should be done by which component in order to reduce energy

Keywords—IEEE1451, TIM, Standard, Sensor Network, Vehicular Sensor Network, VSN.

I. INTRODUCTION

The Information and Communication Technologies (ICT) such as the Internet have enabled people and devices to communicate with each other. At the same, and due to the urge of smart application and smart devices (things), more and more devices are connected to the Internet and, ultimately, connected to each other as well, it is reached the concept of Internet of Things (IoT).

IoT is commonly associated with Sensor Networks (SN), where sensor nodes communicate their sensed data to a user network or to the Internet. SNs are commonly implemented to monitoring some physical dimension. With the integration of actuators in SNs, it is possible to develop control and monitoring systems. On the other hand, SNs implemented on vehicles to monitor urban environment, enable them to communicate in three different ways: Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), and Vehicle-to-Device (V2X) [1]. The integration of IoT technology in any, or all of the three communicating paradigms to a transportation system is known as Vehicular Sensor Networks (VSN) turns it into a connected and an Intelligent Transportation System (ITS) [2]. VSN have been implemented in transport systems to perform monitoring activities [3-4], and other activities beyond that [5]. Authors

that have been showing proposals for full VSN architectures are not focus on the technologies to use in all levels of development a Wireless SN (WSN) [3] nor are using standards from IEEE 1451 family [4], and thus they are not focus on foreseeing issues on future VSN evolution.

As the concern about the negative effects of air pollution is increasing due to values researchers have been getting from the environment [6], the awareness of the governments and population to reduce their footprint grows. One of the solutions being implemented on urban areas is greener transport system solutions [7]. Bike sharing systems are becoming attractive in many European cities, promoting the shift from fossil fuel combustion engine vehicles to a more sustainable and healthy way of transportation [8].

In a world where users search for devices and applications every time smarter, the IoT has a great potential and can be applied in a variety of applications. Although, the IoT concept is promising, it has been growing so fast that the standardization is not following at the same pace [9].

Academic researchers and industry have been working with IEEE 1451 family of standards, discussing about its potential on industry [10], showing the added value to green transportation systems [5], and the advantages of it when implemented in monitoring systems [11].

The standards from IEEE 1451 are well describing documents that define the architecture, the services, and the methods to be implemented in a transducer's network – network composed by sensors and actuators [12-13]. The main objective of these standards is to promote the adaptability, changeability, scalability and interoperability between the things in a global IoT sensor networks.

The objective of the present work is to describe the most important features to be implemented in order to develop a sustainable VSN compliant with the IEEE 1451 family of standards. It is presented the overall VSN architecture and the devices that compose the network, and the standard services and methods to implement that ensure the minimum functioning of a standardized VSN.

This paper is organized as follows: Section I – Introduction – makes the frame of the issue to be addressed, its context and the objective of the present work. Section II - General Overview of the Network Architecture – presents the

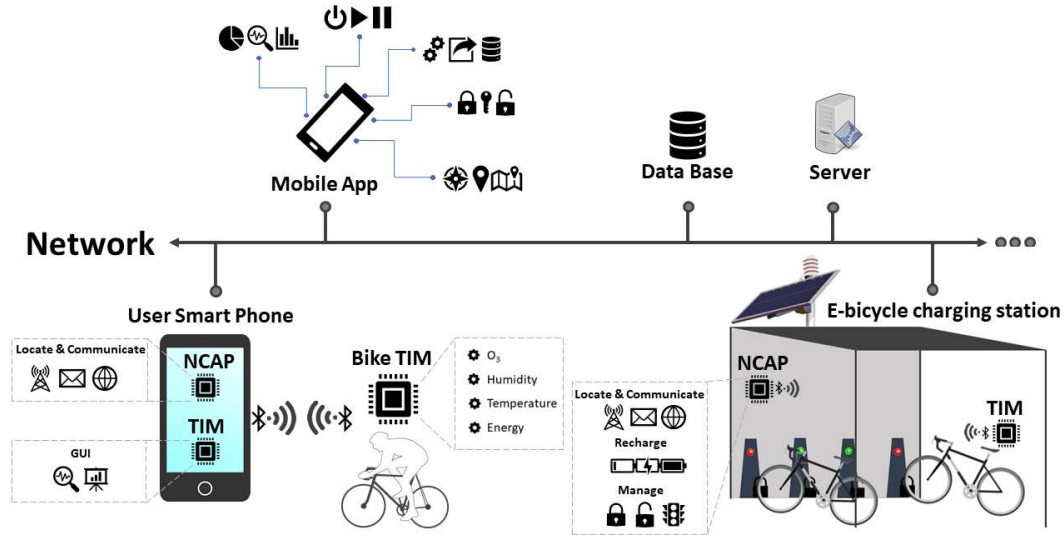


Figure 1. Vehicular Sensor Network Architecture Compliant with IEEE 1451 family of Standards.

architecture and the relationship among the devices that support and integrate the standardized VSN. Section III – Network Elements – details all the elements implemented of in the network. Section IV – Services and Methods –presents information about the services and methods described in the standard that have already implemented in the VSN. The final section – Conclusions – will conclude the paper, providing remarks of the implemented system and considerations for future work.

II. GENERAL OVERVIEW OF THE NETWORK ARCHITECTURE

The objective of developing this VSN is to monitor data such as battery energy usage/consumption, vehicle location, and environmental parameters such as Ozone (O_3), temperature, and humidity. To be compliant with IEEE 1451 family of standards, this VSN shall include a Network Capable Application Processor (NCAP), a Transducer Interface Module (TIM), specific Technical Electronic Data Sheets (TEDS), as described by the IEEE 1451.0 standard [11], and communications protocols described and supported by IEEE 1451.5 standard [14], for wireless communication that do not use RFID (Radio Frequency IDentification) technology.

The present VSN is composed by 50 electric bicycles (e-bike) and 4 Photovoltaic (PV) power stations. Each e-bike has an on-board TIM, which in turn contains all the sensors, with exception to location, making possible to manage e-bike state and process generated data collected.

The e-bike TIM will communicate with a user application, through the bicycle user's smart phone. The communication between the TIM and the NCAP is ensured by the Bluetooth protocol. The NCAP communicates with user application installed in the smart phone. The user location is ensured by the integrated smart phone GPS capability. The architecture of this VSN is represented in Figure 1. The data from sensors travel through the NCAP and user application to a data base, where is permanently stored. The user application is responsible to process stored data and data visualization. A virtual TIM is developed in user's smart phone which

provides the Graphical User Interface (GUI). TEDS are stored in the TIM and will allow the TIM, and all implemented transducers of the VSN, to be self-described, and ensuring standard's architecture. The services and methods to implement shall correspond to those described in IEEE 1451.0, enabling the understanding of TIM's message content. The user application is, in this context, a very import part of the VSN, once it will manage data from sensors to the data base, will manage the user access to the bike-sharing system and its usage responsibility, will guarantee that user/bicycle location, will process data and allow it to be represented in the GUI, and most important, manage the operation of transducers in the TIM's bicycle.

The e-bike TIM is associated to the NCAP of the PV-charging station while parked, or to a user's smartphone integrated NCAP, after being requested. This feature allows to know the e-bike location through smartphone GPS, at least the user location, making him responsible for the e-bike, promoting at the same time a conscious usage of the resource. The operating modes of the VSN are described in [14-15].

The technical schema of the VSN management is shown in Fig. 2. The technical features of the developed TIM for this VSN are described in the following sections.

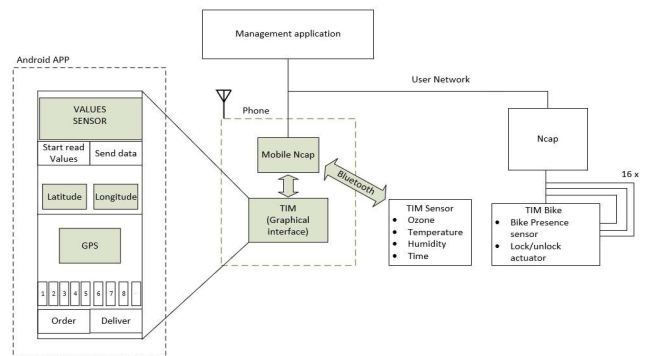


Figure 2. VSN management schema.

III. NETWORK ELEMENTS

The focus of this work is to describe the implementation features of a TIM in a VSN. The TIM microprocessor board implements six transducers: four physical sensors, an embed event-sensor and an actuator. These last two are developed to help calculating the energy being used by the e-bike battery.

A. PV- Station NCAP

The PV stations are sustainable both in terms of energy sufficiency and environment friendly, due to solar energy production system to feed e-bike's battery and low CO₂ emissions, respectively. Each PV station has its own integrated NCAP in order to access both to the transducers implemented on the station and to the e-bike's TIMs, while parked. The implemented transducers will allow:

- Power supply monitoring both in PV generation panels and e-bike's battery;
- Ensure security against misuse or swipe attempt;
- User access management, by enable or disable e-bike's lock;
- Signalling by activated identification LEDs.

Management activities can be either of energy management or fleet management without making any changes to the architecture, once the data analysis is done in a decentralized way on user's application process unit.

B. Mobile NCAP

The mobile application running on the user's smart phone enables an NCAP to be created to connect with the e-bike on-board TIM, provided by the application after the user's request, and allows user to interact with the VSN through a GUI, as shown in Fig. 3. At the same time, smart phone's resources are turned into accessible virtual TIMs through NCAP to be part of the network, such as the GPS capability or time related information. It is through the smart phone

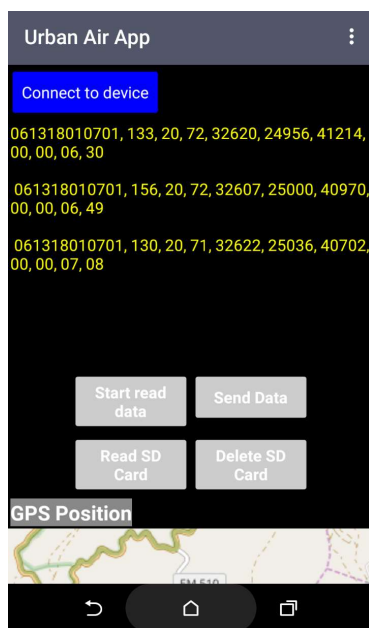


Figure 3. Mobile Application GUI.

application that a user can request from and return to the PV-charging station.

The user application implemented on the smart phone enables two types of communication: Physical TIMs, from e-bikes, connected through the Bluetooth interface; and Virtual TIMs connected through a socket. Although this approach is not described in the IEEE 1451 family of standards, it is intended that its implementation helps to understand the standard concepts related to software development.

C. Virtual TIM

The interface user application on the user smart phone allows the user to receive and visualize information through buttons or dialogue boxes, and status markers. In this context, it is possible to lock to the first ones as sensors and to the second ones as actuators, and at the same time, bring to the software layer these objects – sensors and actuators.

Buttons are developed as a digital sensor. Every time a button is pressed on, it sends a message to the server with the notification of the button status. The swiper is developed as an analogue sensor. Is intended to set positions and the location is read by buttons request. The marker is developed as an actuator. The text it receives is updated in the output. The GPS location is developed as an analogue sensor. There is an object that uses the smart phone GPS capability when it receives a trigger, then the position is sent. The GUI enable data and information visualization, as shown in Fig. 4.

If each object is associated as transducer channel, its integration in the interface can be done almost directly.

D. e-Bike TIM

The TIM implemented on the e-bike allows environmental data to be collected: O₃, temperature, humidity, as well as battery's energy consumption.

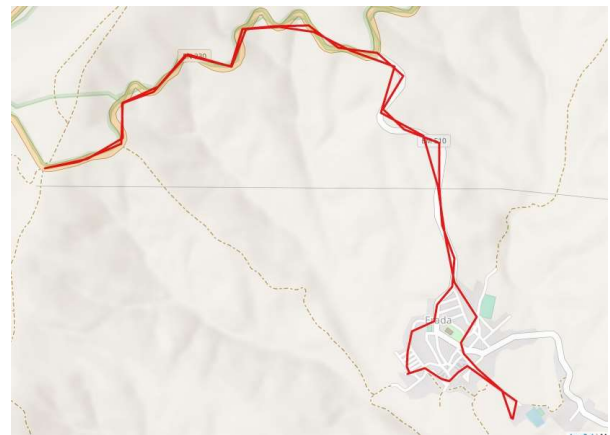


Figure 4. E-bike location information visualization.

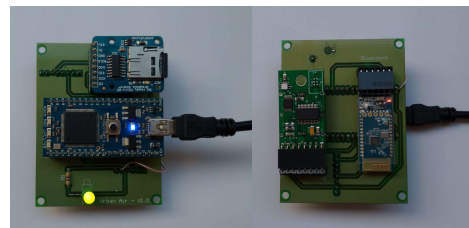


Figure 5. TIM of the VSN.

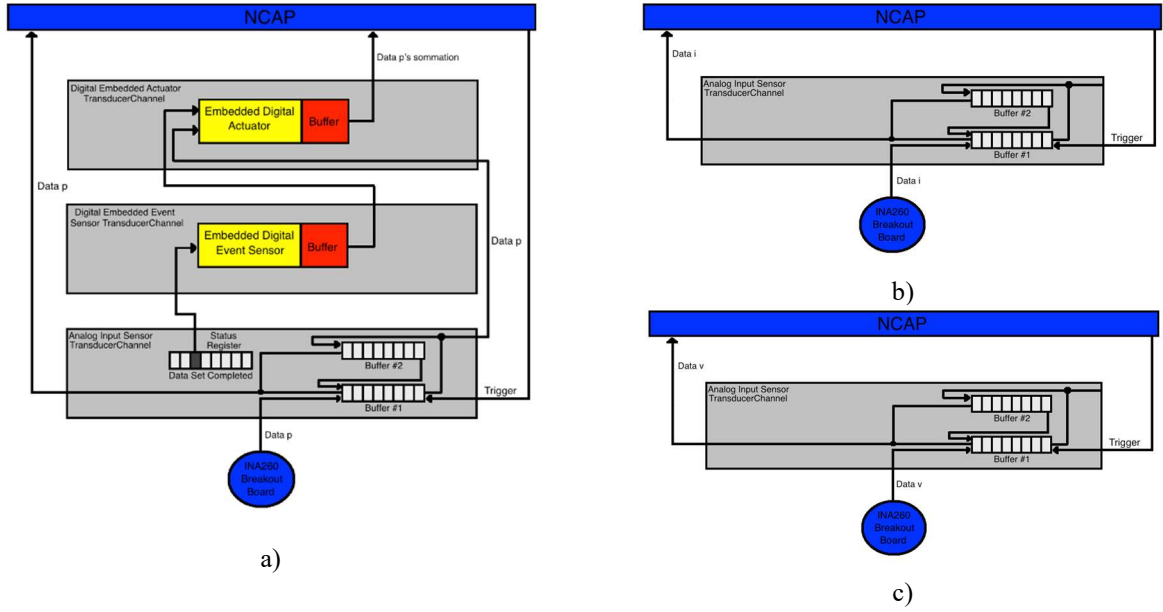


Figure 7. a) Power sensor operating mode with Control Group. b) Current sensor operating mode. c) Voltage sensor operating mode.

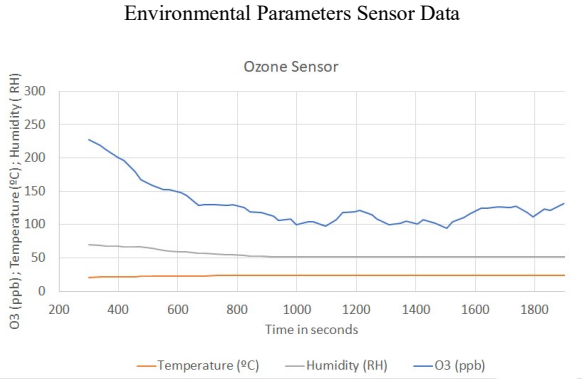


Figure 6. TIM's environmental parameters sensor data visualization.

The e-bike TIM is developed around a mbed NXP LPC1768 module. All sensors are connected to this module. The environmental sensor is from SPEC-Sensors (DGS-O3 968-042) and allow to measure O3, temperature, and humidity. Energy is measured using the INA260. A Bluetooth module connects the TIM with the mobile phone application. A memory card is also available to store information as a backup information system. Board is shown in Fig. 5.

The O3, temperature, humidity and energy sensor generate data that is stored it in data set. This data will be post processed, turned in to information useful to the user, or made available for visualization as shown in Fig. 6. The TIM sends the data in the data set back to the application using “Buffer is Full” mode, one of the modes described in IEEE 1451.0. The first three sensors are very simple: the generated data is directly stored in the specific data set (O3, temperature, humidity), and when the buffer is full, data is sent to the application, through the NCAP, and user application. However, the energy sensor computes the energy, from measured current and voltage values. To know the energy being consumed by the vehicle battery, INA260 continually do the following calculations: first it calculates the power according to (1)

$$P(W) = I(A) \cdot V(V). \quad (1)$$

Then, the energy consumed in the considered time interval is calculated by the TIM according to (2):

$$E(J) = P(W) \cdot \Delta t(s). \quad (2)$$

As the TIM needs to perform these calculations, it is necessary to set up a Control Group as it is shown in Fig. 7 a). The application can also request, for functional, evaluation, current and voltage values, as seen in Fig. 7 b) and Fig. 7 c), respectively.

The operating modes of an implemented Control Group is described in [7]. The energy sensor collects current, voltage, and power data and stores it in the buffer. An event-sensor is implemented in order to set an actuator every time the event occurs. The event is the buffer full: every time it is full, a status register bit of the Status Register is set to 1 and triggers the actuator. In its turn, the actuator gets the data from the buffer and calculates the energy. After, the energy value is sent to the user application. Once the TIM receives a global trigger command, all implemented transducers star to operate at the same time. A Control Group is particularly interesting, because it enables sensors and actuators to operate automatically with the event sensor help. On the other hand, it is possible to trigger only the Control Group and thus prevent all the transducers to functioning without being necessary. A Control Group has a specific address, identified in Meta-TEDS, and so, a trigger can be sent to that address.

IV. SERVICES AND METHODS

The IEEE 1451.0 describe the methods and services to be implemented by the both TIM and NCAP in order to follow standard guideline and make possible the communication between any user application and the sensors. In the Fig. 8 is represented the sequence of services (query and read TEDS, triggering, and read transducers' data set) and methods (request-response) implemented in the TIM.

The “Query Meta-TEDS” and “Query Channel-TEDS” methods have the objective to give the user the size of the

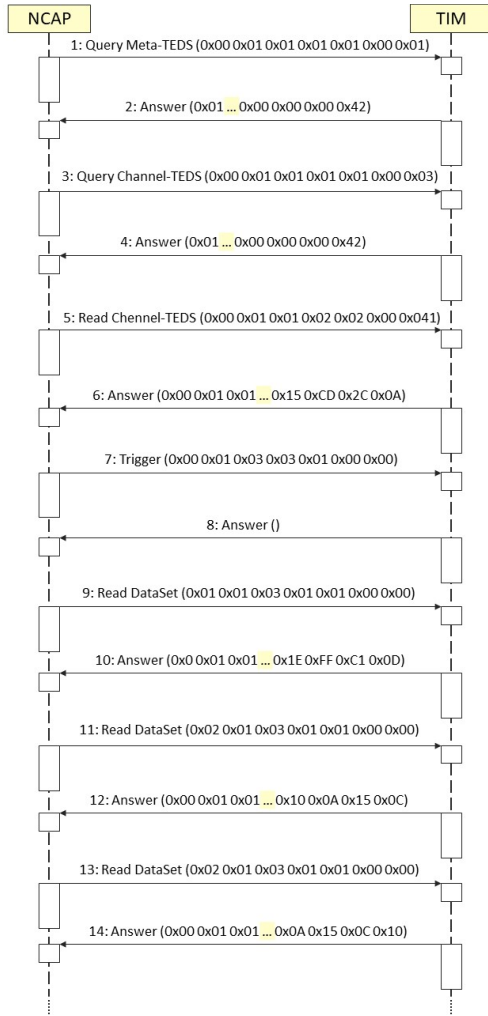


Figure 8. Sequence Diagram of Request-Response of User Application and TIM.

correspondent TEDS in order to be able to build a “Read Meta-TEDS” and “Read Channel-TEDS” request, respectively, once that information is necessary.

The two read methods, “Read Meta-TEDS” and “Read Channel-TEDS”, implemented are very important for the description of TIM and transducers’ features. Meta-TEDS contain information such as how many transducers are implemented, if there is an implemented Control Groups of a set of transducers, data acquisition characteristics, and more. Transducer Channel TEDS contain all the information of corresponding transducer, either if it is physical or embedded.

The trigger command is important to make the TIM’s transducers begin to operate. In this example was implemented the “Trigger Initiated” mode, i.e. the TIM’s transducers start functioning after receiving a trigger command. Also, in this case, the trigger command is global trigger, meaning that all transducers implemented shall all start operating. This ensures that all transducer start functioning at the same time, reduces development time, and reduces energy spent on communication by the TIM, which has energy restrictions. This is the only implemented command that does not require and answer.

Although the Fig. 8 does not show the last example, the next four methods were implemented so the user application can request the reading of the content on sensor data set, where

sensed data is stored before being sent to the data base. The “Read DataSet” method implemented is the same, but the transducer address changes according to the sensor the user application wants to read from. The list of transducers’ addresses is information present in the Meta-TEDS. To know which transducer corresponds to the address, the user needs to read Channel TEDS. Each one has its own Transducer Channel TEDS that contains specific operation information related to the transducer.

The message structure of the request and responses is according to the IEEE 1451.0. For example, the query request to the TIM’s Meta-TEDS has the content shown and described in Table 1. And the correspondent response from the transducer of the e-bike TIM is presented in Table 2.

Table 1. Message structure and content of a “Query Meta-TEDS” request to the TIM.

Destination Most significant	Transducer address = 0	0x00
Destination Most significant	TIM address = 1	0x01
Command Class	Common Class = 1	0x01
Command Function	Query Function = 1	0x01
Length Most Significant	Dependent octets size = 1	0x01
Length Least Significant		0x00
Dependent octets	Access code = 1	0x01

Table 2. Message structure and content of a response from a “Query Meta-TEDS” request.

Success/Fail	TIM address = 1	0x01
Length Most Significant	Dependent octets size = 1	0x01
Length Least Significant		0x00
Dependent octets	TEDS attribute = 0	0x00
	TEDS Status = open to manufacturer	0x07
	TEDS size =66 octets	0x00 0x00 0x00 0x42
	TEDS CheckSum	0xF3 0x6C
	TEDS Max Size =66	0x00 0x00 0x00 0x42

By standard suggestion, the Meta-TEDS shall be read only, and so, the “TEDS attribute” is Zero (0x00) – read only. For the same reason, the “TEDS Max Size” is set to the same size of the actual TEDS, to prevent information to be later added to.

V. CONCLUSIONS

In this paper was presented a VSN based on IEEE 1451 family of standards. It respected the overall architecture and implements services and methods described in IEEE 1451.0.

The focus of this work was showing the implementation of a TIM fully compliant with the standard in a standardized SN. The implementation context is a green and sustainable electric bicycle sharing system, with solar energy generation for battery recharges. Taking advantages of this, it was proposed to implement a SN, and thus a VSN, to monitor urban area conditions, such as O₃, temperature and humidity. It is important to monitor these parameters while promoting e-bike usage in an urban area to understand the impact of its usage on air quality and then assess the advantages of shifting from fossil fuel vehicles to greener options not only for urban

population air quality, but also for their health improvements due to the fomentation of physical and outdoors activities.

At the same time, it was implemented an energy sensor with the objective of measure the energy being consumed by the bicycle battery. As the sensor only measures current and voltage, some calculations must be performed to access energy values. The measurement of battery energy usage will help to describe user's usage profile and mapping the areas where electric bicycle is mostly used.

The interoperability is not an issue in this implementation example. Although, all sensor modules are from the same manufacturer, they can be form diverse manufacturers. The TIM implements standard services and methods that allow the user application access data being generated. Sensed data is stored, temporally in data set within the TIM and in its turns, after a read request, the TIM can build a response answer with sensors' data. It should be point out that, although the focus of this work is the TIM component of the SN, all other levels of VSN architecture are IEEE 1451 compliant, which allows the access to TIM to get data as easy as send a set of request-reply to it following IEEE 1451.0 data type and structure requirements and guidelines.

Processes like calculations the ones to know the energy being consumed are less energetic operations than communication operations. For this reason, the energy is being calculated by the TIM and sent after. This way, useful information can be stored in the data based. Otherwise, current and voltage should be stored and then processed. If by any means, the user needs/wants the current or voltage profiles, it is possible to perform a "Read DataSet" request to the sensor and get current and voltage directly.

VI. FUTURE WORK

For future work it is intended to implement other sensors, such as fall detection. Also, new added sensors to the VSN will be from different manufactures to prove the interoperability between components as well as the reduced effort on adapting new hardware to follow IEEE 1451 family of standards. This, will turn the bicycle smarter, being able to call for help when an unexpected fall occurs.

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